

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

Agents hand each other evidence references, not paraphrases.

The receiver resolves the reference, re-hashes the record, re-reads the source. A signature fixes the record's bytes, not the measurement.

Names move: “the north field” drifts; a 64-bit cell id of about 10 m does not. **Values move:** 0.4709 becomes “about 0.47”; a fact_cid breaks if one bit changes. $\Delta z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$: world, instrument, misregistration, model, noise; change at one pinned address.

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Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)

1 What survives an agent handoff?

Handed prose, agents acted on 254 of 276 corruptions; handed a reference they check, on 0 of 300.

A relay or a faulty signer corrupts one signed NDVI record in one of 16 ways; Agent B resolves, re-hashes, binds, checks, recomputes and re-reads.

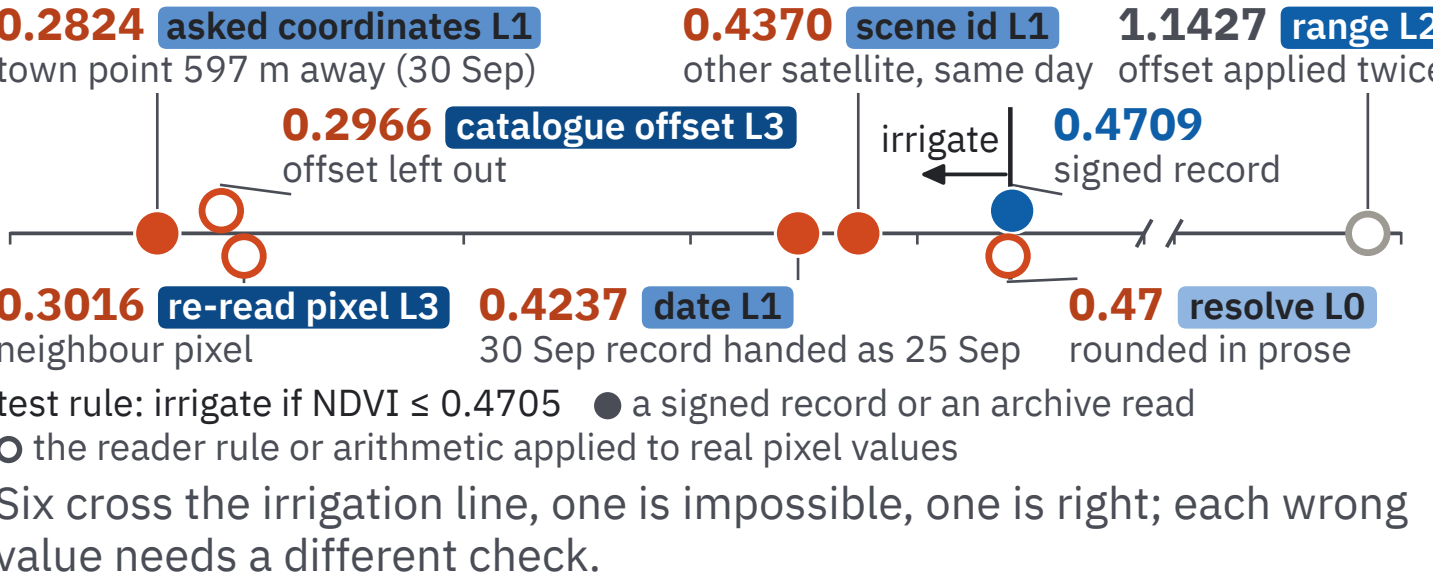


With a model as B and a forged cell, Claude Haiku 4.5 declined 5 of 5; Qwen2.5-3B dropped the cell from all 24 calls and acted 5 of 5: a reference protects only a receiver that keeps it whole and obeys a refusal.

2 What can one NDVI become between agents?

Eight answers; one is right

One question, NDVI at the Keylong field on 25 Sep 2026; real records, archive pixels and handling rules give eight values.



3 Why does an agent handoff need a record?

Our own errors name the checks

Each error happened inside emem, a system built for evidence, or in our tests; each has a published precedent.

"errors may propagate silently across steps" (Munir et al. 2026, arXiv 2604.24919) each rung happened, in emem or in our tests ↑ passes more checks	none: L4; fixed 30 Sep 2026
8 Wrong thing "this image" resolved to a hair salon in Ontario receipt, Merkle proof and state chain all valid toponym ambiguity (Griffa et al. 2018)	re-read only: L3; fixed 28 Sep 2026
7 Wrong pixel, signed the reader rounded the pixel index 162 of 200 sampled records GDAL RFC 33: half-pixel shift; one-pixel misregistration: error > 50 % of NDVI differences (Townshend 1992)	re-read: L3; fixed
6 Missing data signed as 0 off-tile pixels signed as 0 a forest-loss screen passes Hansen lossyear 0 means no loss	re-read: L3; fixed
5 Wrong scale or offset WorldPop signed per pixel, not per km²: 1.77x low one pixel: 0.4709, 0.2966, 1.1427 under three offset rules Element84 items say "offset applied" and "apply -0.1"	recompute: L2; catalogue: L3; fixed
4 Wrong version 918.0 m, then 915.07 m, one band name an earlier citation looks wrong GFC2020 V3 cut forest cover by more than 20 % in the Cerrado	source + as-of: L1; both kept
3 Wrong place coordinates given, town point answered 597 m; NDVI 0.28 for the field's 0.42 GDAL 3 follows CRS axis order (RFC 73)	asked coordinates: L1; fix unconfirmed
2 Wrong date asked 23 Sep, served 25 Sep 10 of 10 agents saw one scene twice; 3 said no 23 Sep scene existed STAC item search defines no default order	bind the date: L1; open
1 Lost in the handoff 0.47 for 0.4709 5 of 5 receivers irrigated Perez et al., ICLR 2025: LLM transmission chains drift	resolve the reference: L0; test rule

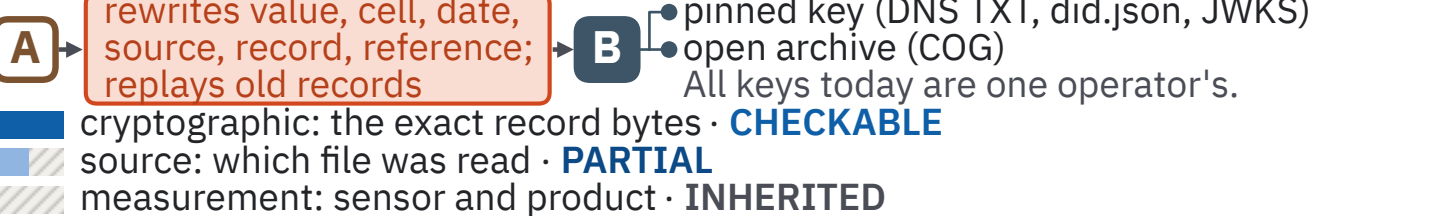
Signatures prevented none; the record made each checkable at a named layer, except the top row.
A check protects only if it runs: emem's resolver once reported a cell match it never tested; one framework adapter returns a refusal as plain text; MAST: no or incomplete verification (FM-3.2).

We ask four questions

- RQ1** Can a receiver detect corruption without trusting the sender? → 1, 5
 - RQ2** Which corruption classes can it detect? → 5
 - RQ3** Does a reference beat prose, JSON, retrieval or an opaque id? → 1, 5
 - RQ4** What stays unverifiable with the reference intact? → 6, 7
- H1** A checked reference refuses corruptions that keep the stated value.
H2 Only a source re-read catches a signer's wrong pixel. **H3** A historical reference keeps the cited state.

A relay can rewrite, not sign

A relay can rewrite what it carries but cannot sign under the pinned key or match an address; a signer's misread pixel is caught only by a re-read. Out of scope: a compromised key, a wrong sensor, the entity meant, the decision.



12 One evidence protocol, multiple agent runtimes.

One reference resolved to one address and one value through 11 client paths, with receipt signatures checked on 9 (30 Sep 2026). Agent hosts: emem is a Claude Code plugin (marketplace emem@emem 2.4.2, 19 skills, 3,672 tokens a session: LIVE, installed by us) and a ChatGPT app (@emem: listed by its publisher, not re-checked by us). A listing is not an integration. Not run by us: ChatGPT, Claude.ai, Dify, VS Code, Cursor.



AGENT HOSTS	INTEROPERABILITY	DISCOVERY	DEVELOPER	FRAMEWORKS
● Claude Code ● Claude Code plugin ● Claude.ai custom connector ● Dify Marketplace plugin (community)	○ Model Context Protocol (MCP) emem.dev/mcp ○ Agent2Agent (A2A) signed agent card	□ Official MCP Registry io.github.vortex-ai/emem □ GitHub MCP Registry ● github.com/Vortex-AI/emem □ Glama	● Gemini CLI ● Visual Studio Code ○ Cursor ● pip install ememdev ● npm i @vortexai/emem ○ REST / OpenAPI 3.1 ● ghcr.io/vortex-ai/emem	△ LlamaIndex △ AutoGen △ CrewAI △ Mastra

● LIVE connected or published
○ PROTOCOL open surface, client not run by us
□ REGISTRY a listing
△ EXAMPLE repo code, tool list checked
checked 30 Sep 2026
ChatGPT (@emem): listed by its publisher, not confirmed by us; held off the band

Mechanisms read from emem's code; where its docs differ, the code is printed. emem.dev at commit 8e9b401; measurements 29 Sep to 1 Oct 2026. Extending verification to execution: reference harness, no spacecraft enrolled. · Same pattern in software supply chains: Sigstore, RFC 9162, SCITT (RFC 9943). Closest agent memory: ARC (arXiv 2607.25066). · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · Reg. (EU) 2023/1115 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Dang et al. 2026, ARC (2607.25066) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)



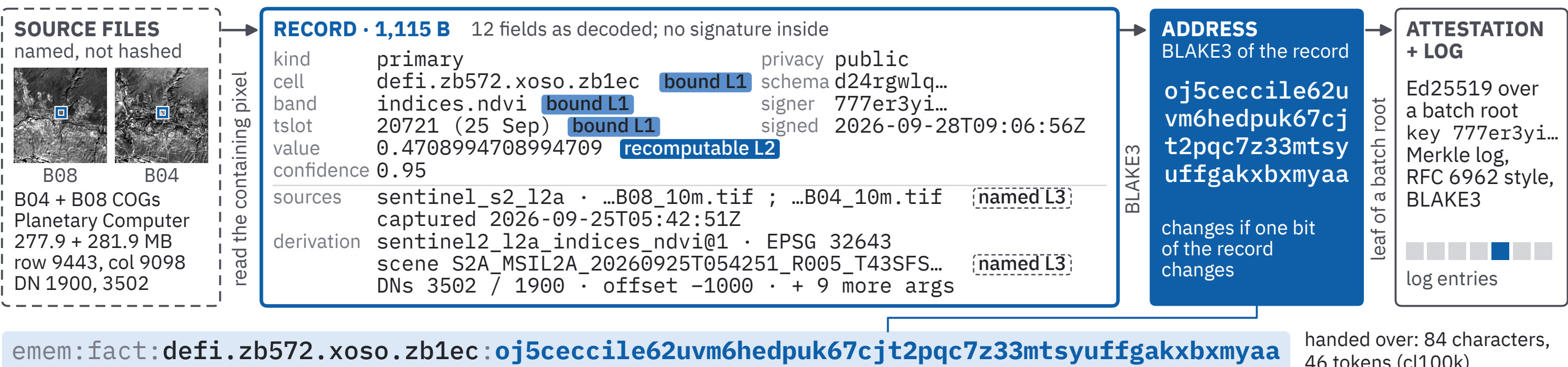
The record this board follows: NDVI 0.4709
Sentinel-2A L2A · Keylong, Lahaul, India · 25 Sep 2026
true colour B04/B03/B02 from signed 10 m grids; linear 1 to 99 % stretch, y 1/1.35

4 What exactly is handed over?

The address names the record, not the pixel.

The receiver gets 84 characters; everything else it fetches and checks.

INSPECT THE RECORD The 1,115 bytes behind the main example, decoded.



The address is the BLAKE3 hash of the 1,115-byte record, which names its source files and the point it read but holds no hash of them; a batch signature covers such addresses. Re-signing makes a new record: one Bengaluru value has seven.

5 Which check stops which corruption?

Each of five checks alone stops a corruption.

agents as B: three Claude models pooled, 1 Oct 2026	prose	JSON	RAG	opaque id	EMEM, full check	first check that refuses
● B acted on corrupted evidence ■ seen in production □ the only check that stops it ● flips the decision						
G0 nothing altered: accepted everywhere, never refused						
M1 stated value moved by 1 ULP	12/12	12/12	12/12	9/12		C L0
M2 stated value rounded to "0.47"	6/12	6/12	11/12	0/12		C L0
M8 value 0.45, re-encoded and re-hashed	12/12	6/12	12/12	6/12		F L0
M9 record cited for another cell	7/12	0/12	6/12	6/12		E L1
M10 cell changed inside the record, re-hashed	7/12	11/12	6/12	6/12		F L0
M11 tsolt changed to look current, re-hashed	12/12	6/12	6/12	6/12		F L0
M12 a record for another band handed over	12/12	0/12	6/12	2/12		E L1
M11 source scene id changed, re-hashed	12/12	12/12	12/12	12/12		F L0
M12 offset changed, value recomputed, re-hashed value (0.46) disagrees with the signed DNS	12/12	11/12	12/12	8/12		F L0
M14 an older record handed over as current	12/12	1/12	6/12	1/12		E L1
M16 a second signed version shown only to B	12/12	6/12	6/12	6/12		G L0
M17 1 ULP changed inside the served bytes	12/12	12/12	12/12	12/12		D L0
M3 token miscopied by one character	n/a	n/a	n/a	0/12		C L0
M13 value 0.45 under the forger's own key	12/12	6/12	12/12	6/12		F L0
M15 DNS read from the pixel 10 m south	12/12	12/12	12/12	12/12		I L3
entity M17 same record; A meant another entity (out of scope)	not run	not run	not run	not run		none

B acts on corrupted evidence

● resolve L0 ● hash L0 ● binding L1 ● signature L0 ● log L0 ● recompute L2 ● re-read L3
B: agents' false acceptance, 1 of n per cell, n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5 runs); squares: the deterministic ceiling. Totals pool 23 items, 1 Oct 2026. Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

Binding stops cell, time and band swaps; the signature, re-hashed forgeries; the log, a version shown only to B; recomputing, the signer's arithmetic; only the source re-read, the signer's wrong pixel. The entity meant passes every check.

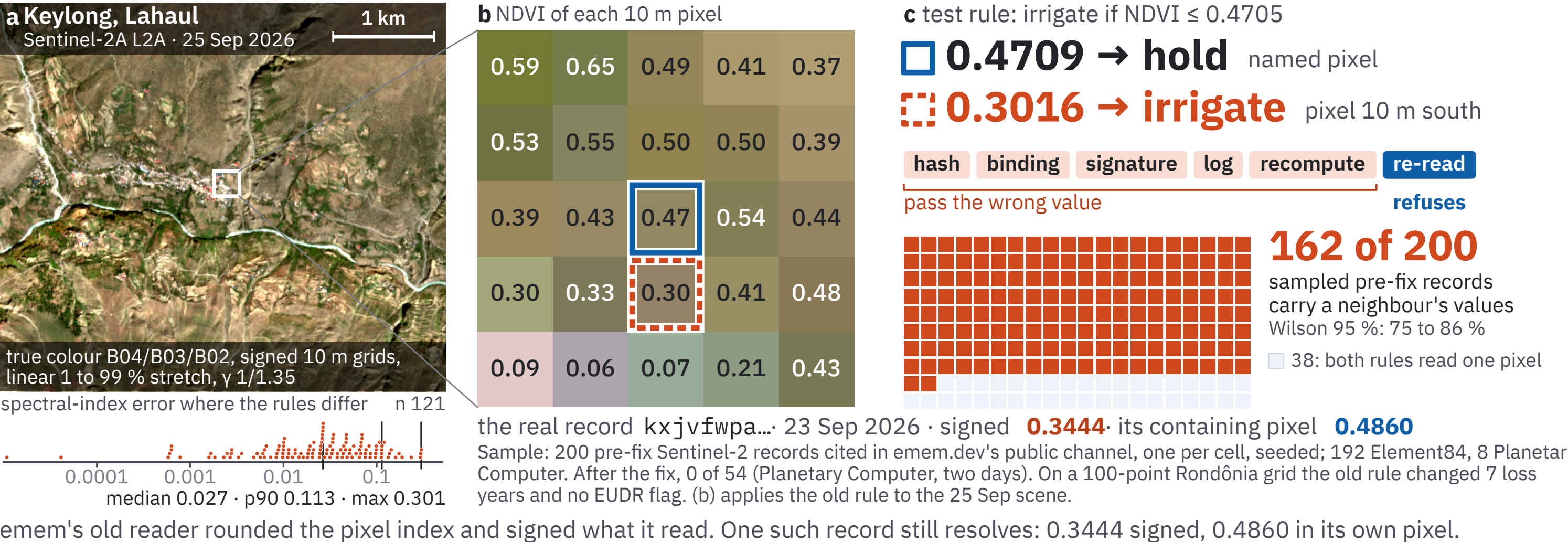
JSON and an opaque id caught about half by reading their fields (112 of 276 and 134 of 288). Haiku with the tool but no instruction acted on 2 of 150, both M24, bound to the handoff's cell, not its question. Live, agents acted on the pre-fix record 4 of 4 uninstruicted, 2 of 4 to resolve and bind, 0 of 4 with a source re-read. 771 rows with no model output (a CLI rate limit) were excluded and re-run; the second model family is Qwen alone.

RE-RUN THE TEST Data and scripts behind this board's numbers, with one re-run command.

6 Can a valid signature carry a wrong measurement?

The right record, the wrong pixel.

A valid signature can faithfully preserve a wrong measurement when the reader itself selected the wrong source pixel.



emem's old reader rounded the pixel index and signed what it read. One such record still resolves: 0.3444 signed, 0.4860 in its own pixel.

DISCOVER INTEGRATIONS

Where EMEM runs today, each surface labelled by evidence.

one address, one value on every path
median ms of 3 runs, log scale, 30 Sep 2026

official MCP Python SDK ✓ official MCP TypeScript SDK ✓ TypeScript SDK ✓ Python SDK ✓ raw REST ✓ raw MCP ✓ A2A message/send ✓ independent BLAKE3 + CBOR reader - LlamaIndex - LangChain MCP adapters ✓ A to B, two processes ✓ receipt signature checked on 9 of 11	56.3 58.3 58.6 207.7 223.5 223.7 226.3 235.6 300.8 772.8 · one run 1,176.8 1,000
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13 EMEM relies on these layers

A STAC item identifies a file; EMEM identifies the observation an agent cited, for the next agent to re-check. GeoGuard judges the claim; EMEM fixes the evidence it cites.

JUDGE	GeoGuard	claim in text	resolve · re-hash · re-read
CARRY	MCP, A2A	a tool call, an agent card	hand off
RETRIEVE	RAG	a text chunk	
SIGN FILES	C2PA, CDSE Traceability	a file	EMEM: is this the observation the sender cited?
RECORD LINEAGE	W3C PROV	entity, activity, agent	cite
RUN	openEO	a process graph	
FIND	STAC	an asset (file)	cite

A signed observation reference left the receiver acting on none of the 16 corruptions in our suite, and agents with a checked reference acted on 0 of 300; entity, sensor and decision stay beyond its checks.
READ THE METHODS vortex-ai.github.io/esa-poster
Definitions, denominators, models, and what our tools generated.

